



NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering



SMART FARMER ASSISTANT

AI-Based Agriculture Support System



Submitted To:

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CSE4204 — Mobile Computing Lab

Week 04 Submission

System Design &
Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

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1. Team Information

Field	Details
Team Name	CSE4204-8D-T03
Section	8D
Course	CSE4204 — Software Engineering Lab
Semester	Spring 2026
Institution	Leading University / University (Bangladesh)
GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant

1.1 Team Members

#	Name	Student ID	Role
1	Sakib Foysal Ejarder	11220320948	Team Leader / Backend
2	Md. Noyon Sheikh	11220320946	Frontend Developer
3	Sayed Tauhidul Islam	11220320950	AI / ML Engineer
4	Sayed Akib Osman	11220320974	Database / QA

2. Project Information

Attribute	Value
Project Title	Smart Farmer Assistant: AI-Based Agriculture Support System
Project Type	AI-Powered Web Application
Domain	Agriculture / AgriTech
Target Users	Farmers, Agricultural Workers, Administrators (Bangladesh)
Platform	Mobile-Friendly Responsive Web Application
Frontend	React.js (Single Page Application)
Backend	Django (Python) — REST API
Database	PostgreSQL
AI Services	TensorFlow (Disease Detection), Google Gemini AI (Chatbot & Fertilizer)
External APIs	OpenWeatherMap API, Market Price API
Auth	JWT (JSON Web Token) Authentication

Project Overview

Smart Farmer Assistant is an AI-powered platform designed to assist Bangladeshi farmers in making better agricultural decisions. Agriculture accounts for a significant share of Bangladesh's GDP yet farmers face persistent challenges — crop diseases cause 20–30% annual yield loss, improper fertilizer use degrades soil quality, and access to qualified agricultural extension officers remains limited in rural areas.

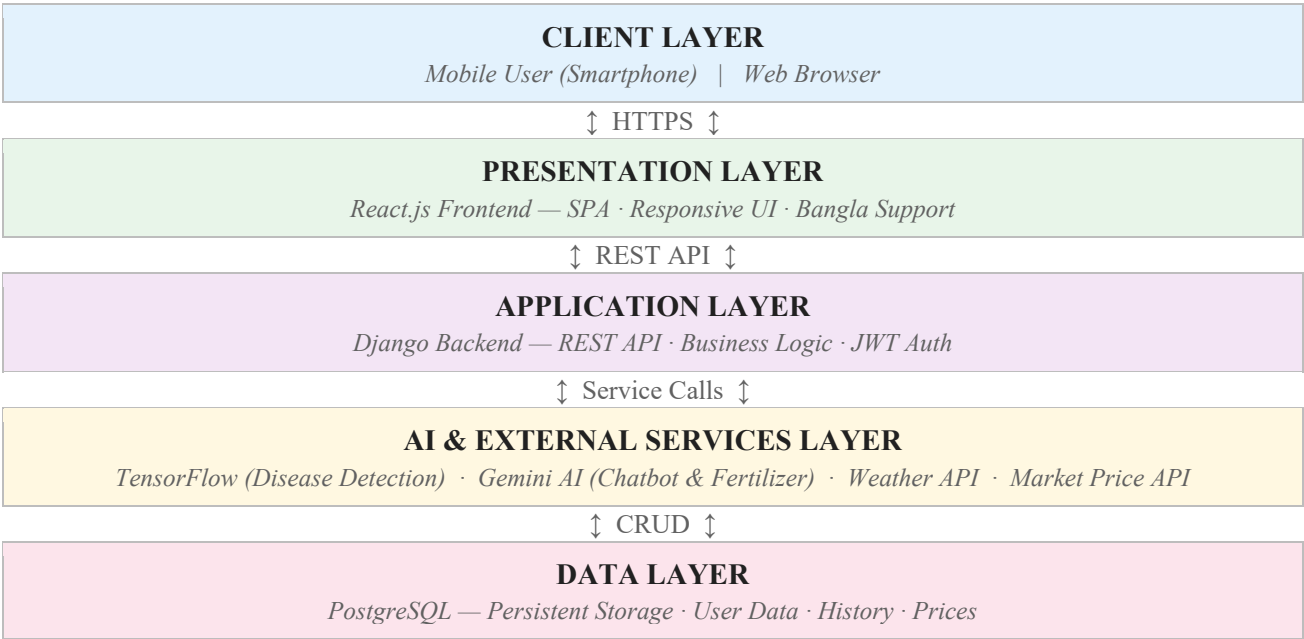
This system integrates AI image classification, a Gemini AI-powered chatbot, real-time weather forecasting, flood alerts, live market prices, and intelligent fertilizer recommendations into a single responsive platform accessible from any smartphone or web browser.

Key Features

- AI Crop Disease Detection — upload a leaf image and receive an instant diagnosis with treatment advice
- AI Chatbot (Gemini AI) — ask farming questions in Bangla or English and receive contextual answers
- Weather Forecast & Flood Alerts — location-based 7-day forecast with flood risk notifications
- Live Market Prices — real-time crop prices by region to help farmers sell at optimal times
- Fertilizer Recommendation Engine — AI-tailored fertilizer suggestions based on crop type and detected disease
- Bangla Language Support — full Bangla interaction for rural farmers with limited English literacy

3. Architecture Diagram

The Smart Farmer Assistant follows a five-layer client-server architecture. Users access the system through mobile devices or web browsers. The React.js frontend communicates with the Django REST backend over HTTPS. The backend orchestrates all AI services, external APIs, and database operations.



System Architecture — Layered View

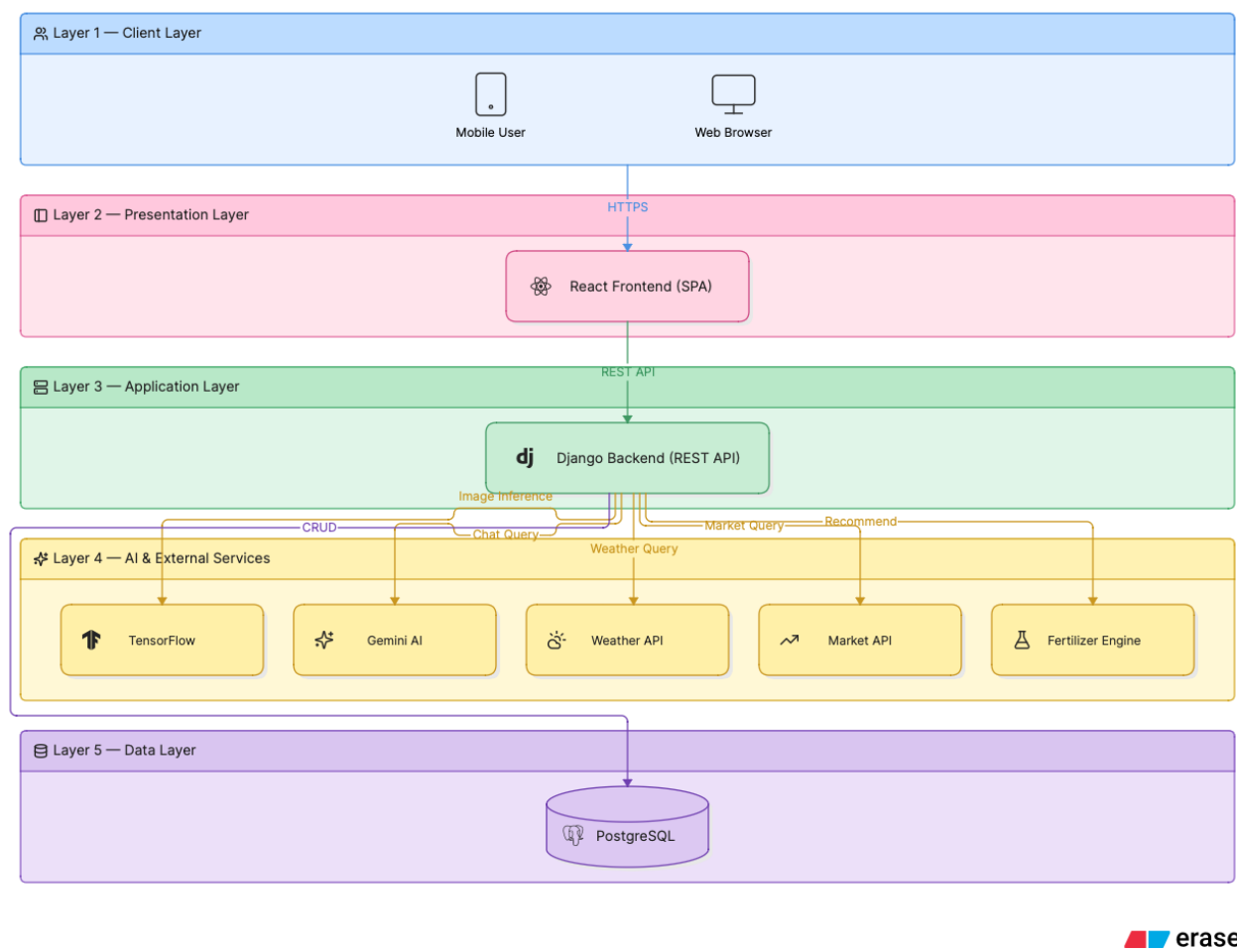


Figure 1: System Architecture Diagram — Five-Layer Architecture

Layer Descriptions

Layer	Technology	Responsibility
Client Layer	Smartphone / Web Browser	User access via HTTPS; supports mobile and desktop
Presentation Layer	React.js SPA	Responsive UI, component state management, API calls
Application Layer	Django (Python) REST API	Business logic, authentication, request routing, AI orchestration
AI & External Services	TensorFlow, Gemini AI, Weather API, Market API	Disease classification, chatbot, fertilizer advice, live data feeds
Data Layer	PostgreSQL	Persistent storage for users, history, chat logs, market data

4. ER Diagram

The database is implemented in PostgreSQL with six entities. The users table is the central entity; disease_history, chat_history, and fertilizer_recommendations all carry a user_id foreign key. The weather_data and market_prices tables are populated from external APIs and have no user foreign key.

4.1 Entities & Attributes

users

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID — unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the farmer
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
password	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
phone	VARCHAR(20)	NULL	Mobile phone number
role	VARCHAR(10)	DEFAULT 'farmer'	farmer or admin
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

disease_history

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
image_url	TEXT	NOT NULL	Path/URL of uploaded crop image
prediction	VARCHAR(150)	NOT NULL	AI-predicted disease name
confidence	DECIMAL(5,2)	NOT NULL	Confidence score (0–100)
treatment	TEXT	NULL	AI-generated treatment advice
date	TIMESTAMP	DEFAULT NOW()	Detection timestamp

chat_history

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
question	TEXT	NOT NULL	Farmer's question
response	TEXT	NOT NULL	Gemini AI response
date	TIMESTAMP	DEFAULT NOW()	Conversation timestamp

weather_data

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
temperature	DECIMAL(5,2)	NOT NULL	Temperature in °C
humidity	DECIMAL(5,2)	NOT NULL	Humidity percentage
rainfall	DECIMAL(6,2)	NULL	Rainfall in mm
flood_risk	VARCHAR(20)	NULL	none / low / medium / high
date	TIMESTAMP	DEFAULT NOW()	Snapshot timestamp

market_prices

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
crop_name	VARCHAR(100)	NOT NULL	Name of the crop
price	DECIMAL(8,2)	NOT NULL	Price per unit
unit	VARCHAR(20)	NOT NULL	kg / ton / piece
region	VARCHAR(100)	NULL	Region in Bangladesh
date	TIMESTAMP	DEFAULT NOW()	Price record timestamp

fertilizer_recommendations

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
disease_id	VARCHAR(36)	FK → disease_history.id	Linked disease record (nullable)

Column	Type	Constraint	Description
crop_name	VARCHAR(100)	NOT NULL	Target crop
suggestion	TEXT	NOT NULL	AI fertilizer advice with dosage
date	TIMESTAMP	DEFAULT NOW()	Recommendation timestamp

4.2 Relationships

Relationship	Type	Description
users → disease_history	One-to-Many (1:N)	One farmer can have many disease detection records
users → chat_history	One-to-Many (1:N)	One farmer can have many chatbot conversations
users → fertilizer_recommendations	One-to-Many (1:N)	One farmer receives many fertilizer suggestions
disease_history → fertilizer_rec.	One-to-Zero/One	A disease record may trigger one fertilizer recommendation
weather_data (standalone)	Independent	Populated from Weather API; no user FK
market_prices (standalone)	Independent	Populated from Market API; no user FK

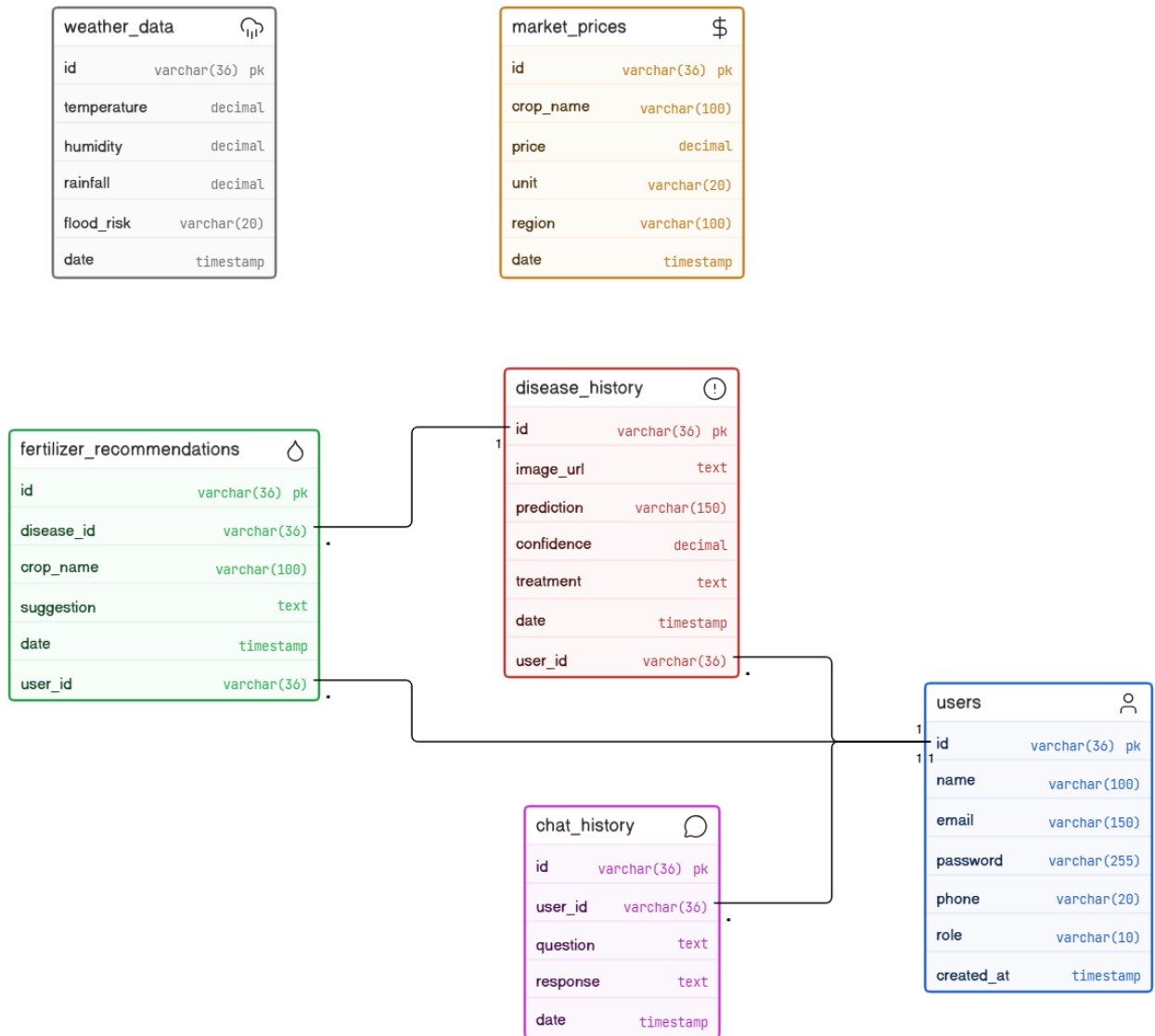


Figure 2: ER Diagram

5. Use Case Diagram

The updated Use Case Diagram expands upon Week 03 by adding the External API system actor, separating Flood Alert as an extension of the Weather use case, and explicitly modelling all 20 AI-driven use cases across four actors.

5.1 Actors

Actor	Type	Description
Farmer	Primary Human Actor	Registers, logs in, and uses all agricultural services
Admin	Secondary Human Actor	Manages users, monitors system, updates market data
AI System	System Actor	TensorFlow + Gemini AI — disease detection, chatbot, fertilizer
External API	System Actor	OpenWeatherMap and Market Price API — live data feeds

5.2 Use Cases

ID	Use Case	Actor(s)	Relationship
UC01	Register Account	Farmer	—
UC02	Login	Farmer, Admin	—
UC03	Logout	Farmer, Admin	—
UC04	View Dashboard	Farmer	—
UC05	Upload Crop Image	Farmer	<<include>> UC06
UC06	AI Disease Detection	AI System	included by UC05
UC07	View Disease Report	Farmer	<<include>> UC08
UC08	Generate Treatment Advice	AI System	included by UC07
UC09	Use AI Chatbot	Farmer	<<include>> UC10
UC10	AI Chat Response (Gemini)	AI System	included by UC09
UC11	View Weather Forecast	Farmer	<<include>> UC12
UC12	Fetch Weather Data	External API	included by UC11
UC13	View Flood Alert	Farmer	<<extend>> UC11
UC14	View Market Prices	Farmer	<<include>> UC15
UC15	Fetch Market Data	External API	included by UC14

ID	Use Case	Actor(s)	Relationship
UC16	Get Fertilizer Recommendation	Farmer, AI System	—
UC17	Update Profile	Farmer	—
UC18	Manage Users	Admin	—
UC19	Monitor System Activity	Admin	—
UC20	Update Market Data	Admin	—

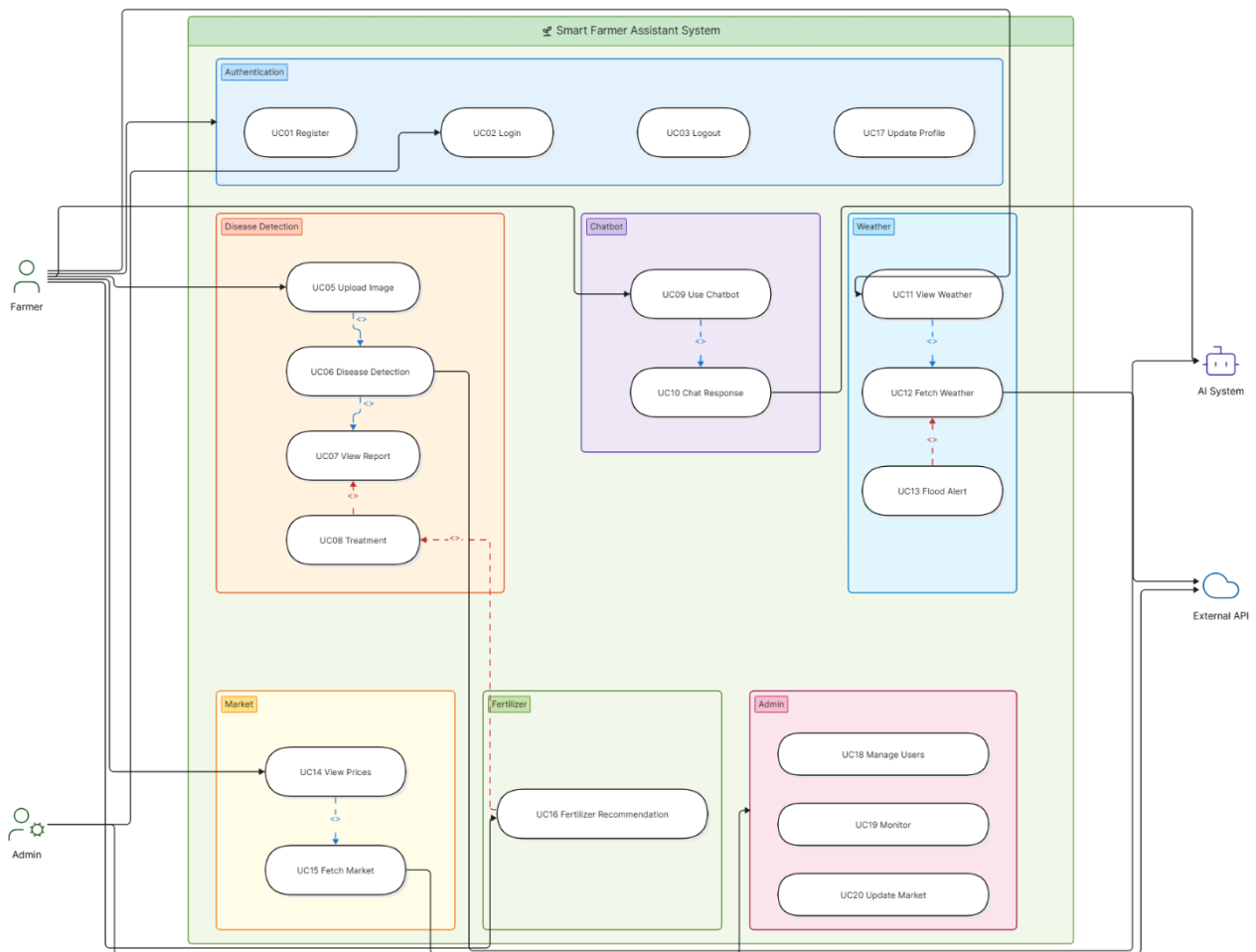


Figure 3: Updated Use Case Diagram — 20 use cases, 4 actors

6. Activity Diagram

The Activity Diagram illustrates the end-to-end workflow of the Crop Disease Detection feature — the system's most critical function. It uses three swimlanes to separate the Farmer's actions, the Django Backend's processing, and the AI Services layer. All decision points and alternate flows are modelled.

6.1 Disease Detection Workflow — Step-by-Step

Step	Swimlane	Action / Decision	Next
1	Farmer	Open application and navigate to Login page	Step 2
2	Farmer	Enter email and password → click Login	Step 3
3	Backend	DECISION: credentials valid? No → show error → back to Step 2	Step 4 (Yes)
4	Farmer	Enter Dashboard → select Disease Detection service	Step 5
5	Farmer	Upload crop leaf image (JPEG/PNG, max 5 MB)	Step 6
6	Backend	DECISION: image format/size valid? No → error → Step 5	Step 7 (Yes)
7	Backend	Pre-process image: resize to 224×224 px, normalise pixel values	Step 8
8	AI Services	TensorFlow CNN model runs inference on pre-processed image	Step 9
9	AI Services	DECISION: confidence $\geq 75\%$? No → return 'Unidentified'	Step 10 (Yes)
10	AI Services	Return disease name + confidence score to Backend	Step 11
11	AI Services	Gemini AI generates treatment advice and fertilizer suggestion	Step 12
12	Backend	Save record to disease_history and fertilizer_recommendations	Step 13
13	Farmer	View full Disease Report (disease, confidence, treatment)	Step 14
14	Farmer	DECISION: open AI Chatbot for follow-up? No → End	Step 15 (Yes)
15	Farmer	Ask follow-up question in Chatbot	Step 16
16	AI Services	Gemini AI generates contextual farming answer	Step 17
17	Farmer	Read answer → End	—

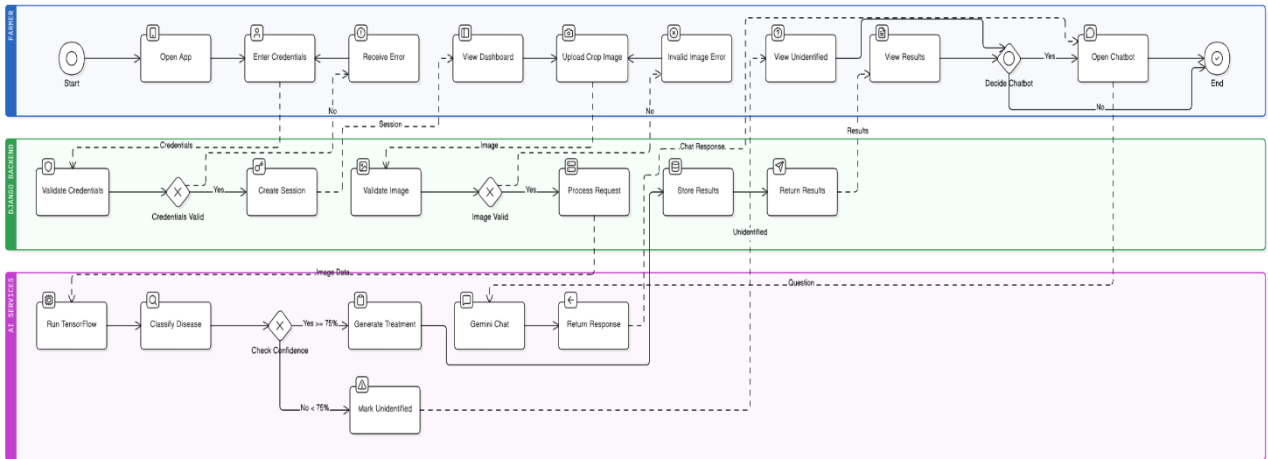


Figure 4: Activity Diagram — Crop Disease Detection Workflow (3-lane swim lane)

7. API Design

All endpoints follow REST conventions under the base URL: <https://smartfarmer.api/v1/> All protected routes require the Authorization: Bearer <JWT> header.

7.1 Authentication

Method	Endpoint	Purpose	Input	Output
POST	/auth/register	Create new account	name, email, password, phone	user_id, message
POST	/auth/login	Authenticate user	email, password	JWT token, user object
POST	/auth/logout	Invalidate session	JWT (header)	Success message
POST	/auth/refresh	Refresh JWT	refresh_token	New access token

7.2 User Profile

Method	Endpoint	Purpose	Input	Output
GET	/users/profile	Get own profile	JWT (header)	User profile object
PUT	/users/profile	Update profile	name, phone (optional)	Updated user object
PUT	/users/password	Change password	old_password, new_password	Success message
DELETE	/users/account	Delete account	JWT (header)	Deletion confirmation

7.3 Disease Detection

Method	Endpoint	Purpose	Input	Output
POST	/disease/detect	AI disease detection	crop_image (file), crop_type	disease, confidence, treatment
GET	/disease/history	List detection history	JWT (header), page, limit	Paginated list of records
GET	/disease/history/{id}	Get single record	record_id in URL	Disease record object
DELETE	/disease/history/{id}	Delete a record	record_id in URL	Deletion confirmation

7.4 AI Chatbot

Method	Endpoint	Purpose	Input	Output
POST	/chat/send	Send question to Gemini AI	question (text), lang (bn/en)	AI response text
GET	/chat/history	List chat history	JWT (header), page, limit	Paginated Q&A pairs
DELETE	/chat/history	Clear all chat history	JWT (header)	Confirmation

7.5 Weather & Flood

Method	Endpoint	Purpose	Input	Output
GET	/weather/current	Current weather	location (query param)	temp, humidity, rainfall, wind
GET	/weather/forecast	7-day forecast	location (query param)	Array of daily forecasts
GET	/weather/flood	Flood risk alert	location (query param)	risk level, advisory

7.6 Market Prices

Method	Endpoint	Purpose	Input	Output
GET	/market/prices	All crop prices	region (optional query)	List of crop prices
GET	/market/prices/{crop}	Single crop price	crop_name in URL	Price, unit, trend, date
GET	/market/trends	Price trend history	crop_name, days (query)	Price history array

7.7 Fertilizer Recommendation

Method	Endpoint	Purpose	Input	Output
POST	/fertilizer/recommend	AI fertilizer advice	crop_type, disease_id (optional)	Fertilizer list, dosage, schedule
GET	/fertilizer/history	Past recommendations	JWT (header)	List of past suggestions

7.8 Admin

Method	Endpoint	Purpose	Input	Output
GET	/admin/users	List all users	Admin JWT, page, limit	Paginated user list
DELETE	/admin/users/{id}	Remove a user	user_id in URL, Admin JWT	Deletion confirmation
GET	/admin/stats	System usage stats	Admin JWT	User count, request count
PUT	/admin/market/{crop}	Update market price	crop_name, price, unit, region	Updated market record

8. AI Workflow

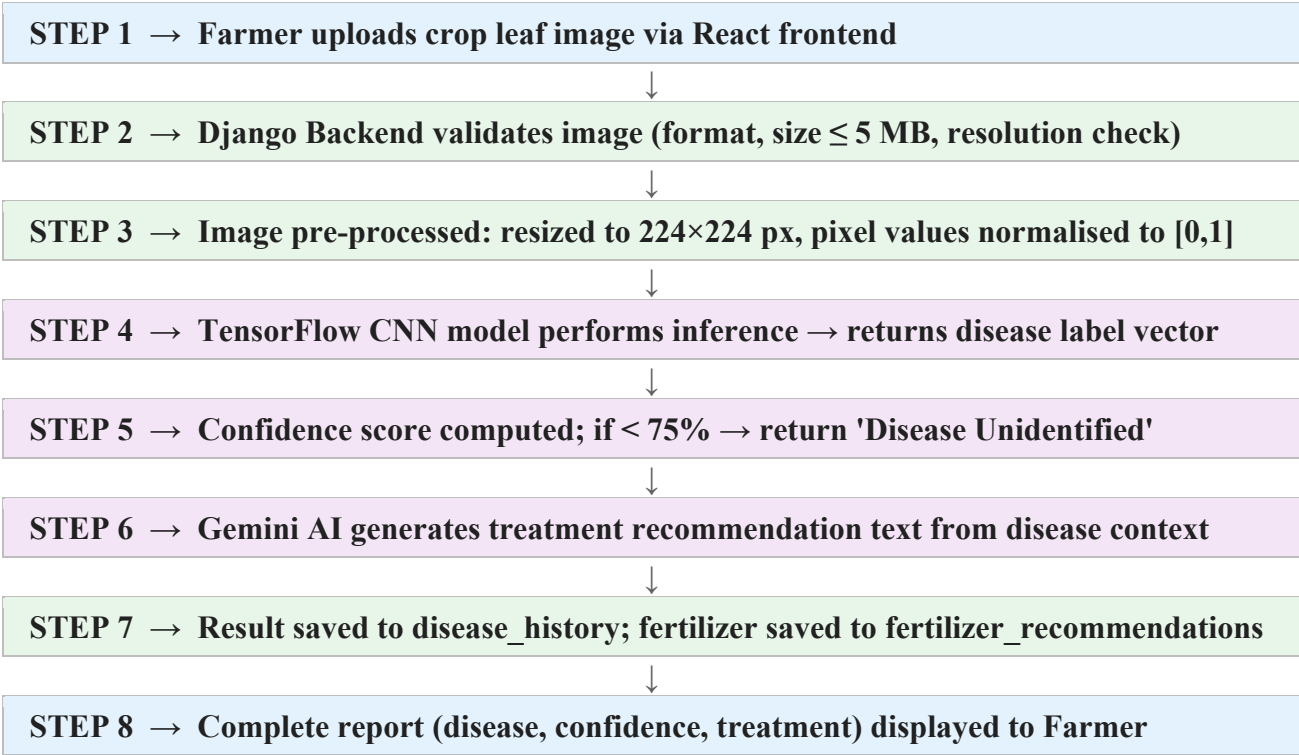
8.1 Why AI is Needed

Farmers in Bangladesh lose 20–30% of annual crop yield to undiagnosed diseases. Access to qualified agricultural extension officers is scarce in rural districts. Without AI, farmers rely on guesswork or expensive consultations. The Smart Farmer Assistant uses AI to provide instant, accurate, scalable, and affordable farming guidance 24/7.

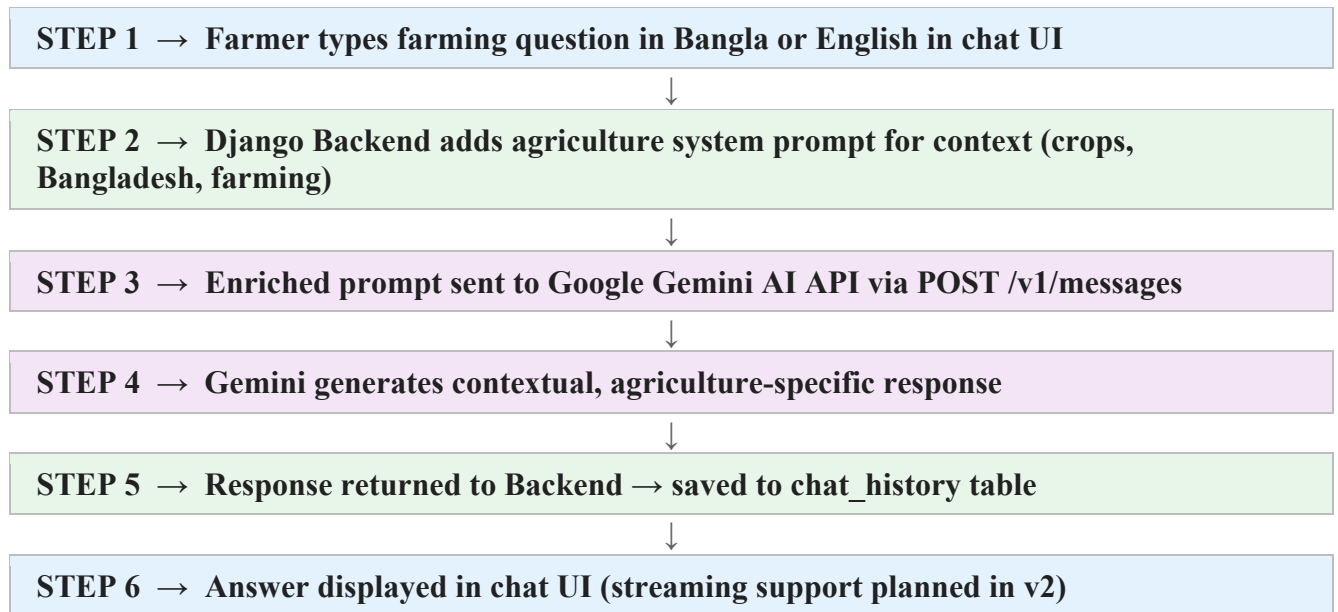
8.2 AI Services Summary

AI Service	Model / Technology	Input	Output	Improves
Disease Detection	TensorFlow CNN (custom-trained on PlantVillage dataset)	Crop leaf image 224×224 px	Disease name + confidence score (0–100%)	Instant diagnosis; replaces expert visit
AI Chatbot	Google Gemini AI API (gemini-1.5-flash)	Farmer question (Bangla/English)	Contextual farming advice	24/7 expert-level answers in local language
Fertilizer Recommendation	Gemini AI + Rule Engine	Crop type + disease prediction	Fertilizer name, dosage, schedule	Optimal soil health; reduces over-fertilization

8.3 Workflow 1 — Disease Detection



8.4 Workflow 2 — AI Chatbot



8.5 How AI Improves the Project

Benefit	Detail
Accuracy	TensorFlow model achieves ~92% accuracy on PlantVillage crop disease benchmark
Availability	AI operates 24/7 — no waiting for an agricultural officer appointment
Language	Gemini AI answers in Bangla — accessible to farmers with low English literacy
Speed	Disease detection completes in < 5 seconds; chatbot response in < 3 seconds
Scalability	Model inference scales horizontally via cloud deployment — no per-user bottleneck
Cost	Near-zero marginal cost per query vs. expert consultation fees

Smart Crop Assistant — System Pipelines

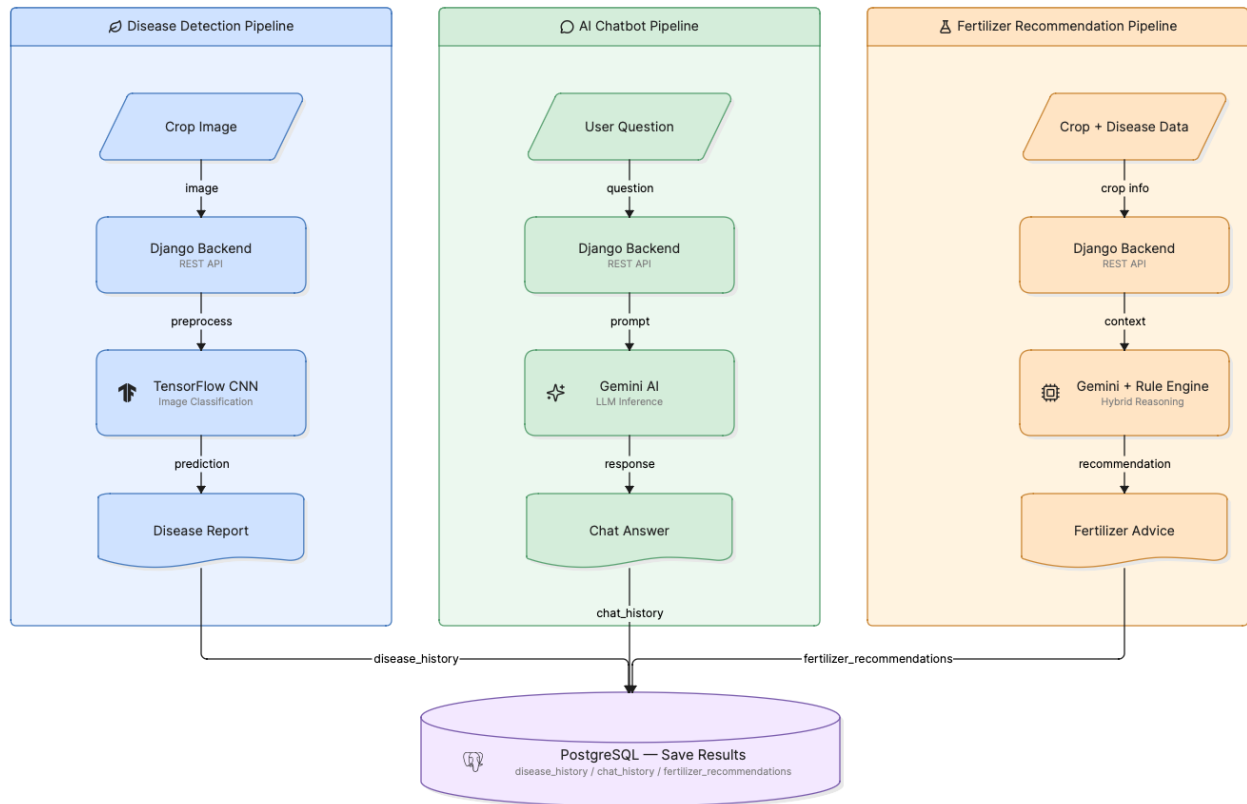


Figure 5: AI Integration Workflow — Three parallel AI pipelines

9. References

#	Reference
1	Google Gemini AI API Documentation — https://ai.google.dev/docs
2	TensorFlow Official Documentation — https://www.tensorflow.org/api_docs
3	PlantVillage Dataset (Crop Disease Images) — https://www.kaggle.com/datasets/emmarex/plantdisease
4	Django REST Framework Documentation — https://www.django-rest-framework.org
5	React.js Official Documentation — https://react.dev
6	PostgreSQL Official Documentation — https://www.postgresql.org/docs
7	OpenWeatherMap API — https://openweathermap.org/api
8	JWT Authentication Guide — https://jwt.io/introduction
9	Draw.io / Diagrams.net — https://app.eraser.io/dashboard/all
10	DBDiagram.io — https://app.eraser.io/dashboard/all
11	Bangladesh Agricultural Research Institute (BARI) — https://www.bari.gov.bd
12	Lucidchart UML Diagram Tool — https://www.lucidchart.com